This is CS50

Favorites

https://docs.google.com/forms/

Guest

Favorites

* Indicates required question

language *

Which is your favorite language?

- Scratch
- C
- Python

favorites - Google Sheets

https://docs.google.com/spreadsheets/

Guest

favorites

File Edit View Insert Format Data Tools Extensions Help

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100% View only

A1 Timestamp

Form_Responses

	A	B	C	D	E
1	Timestamp	language	problem		
2					
3					
4					
5					
6					
7					

Form Responses 1

flat-file database

CSV

key	value

relational database

SQL

C

R

U

D

CREATE

READ

UPDATE

DELETE

CREATE

SELECT

UPDATE

DELETE

CREATE, INSERT

SELECT

UPDATE

DELETE

CREATE, INSERT

SELECT

UPDATE

DELETE, DROP

CREATE, INSERT

SELECT

UPDATE

DELETE, DROP

...

```
CREATE TABLE table (column type, ...);
```

sqlite3

```
sqlite3 FILE
```

```
.mode csv
.import FILE TABLE
.quit
```

```
.schema
```

```
SELECT columns FROM table;
...
```

AVG

COUNT

DISTINCT

LOWER

MAX

MIN

UPPER

...

GROUP BY

LIKE

LIMIT

ORDER BY

WHERE

...

language
C
Python
Python
SQL
Python
SQL

language
C
Python
Python
SQL
Python
SQL

language
C
Python
Python
SQL
Python
SQL

language
C
Python
Python
SQL
Python
SQL

language	COUNT(*)
C	1
Python	3
SQL	2

GROUP BY

LIKE

LIMIT

ORDER BY

WHERE

...

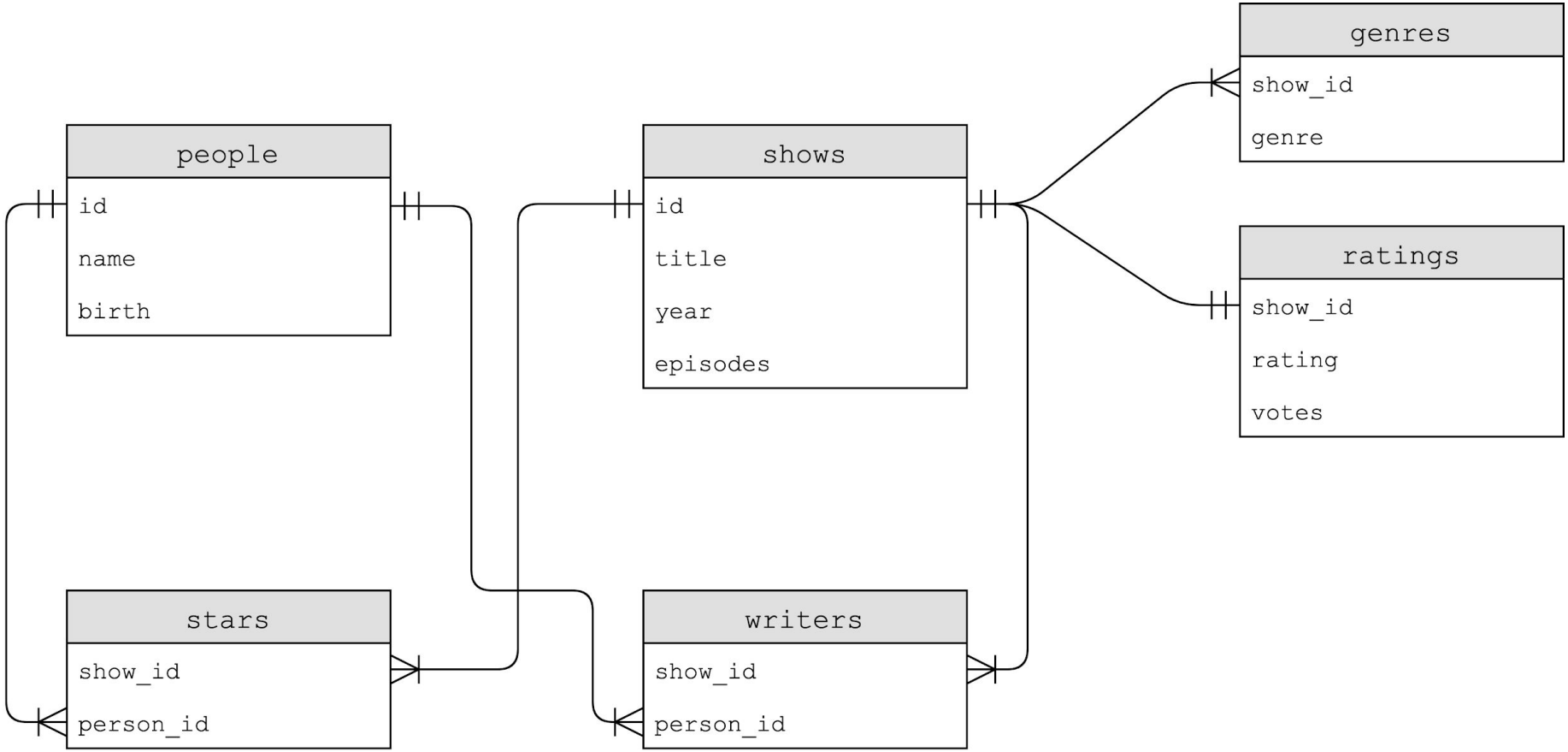
```
INSERT INTO table (column, ...) VALUES(value, ...);
```

NULL

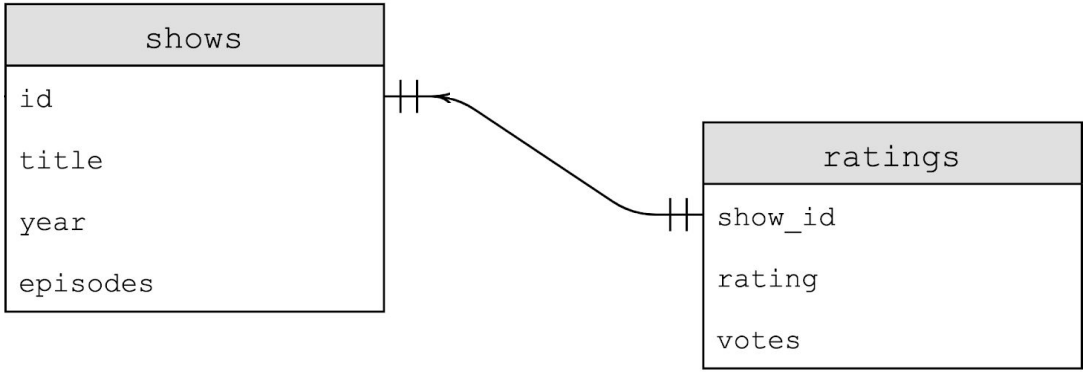
```
DELETE FROM table WHERE condition;
```

```
UPDATE table SET column = value WHERE condition;
```

IMDb


people
id
name
birth
shows
id
title
year
episodes
genres
show_id
genre
ratings
show_id
rating
votes
stars
show_id
person_id
writers
show_id
person_id

shows
id
title
year
episodes


shows
id
title
year
episodes
ratings
show_id
rating
votes

one-to-one

BLOB

INTEGER

NUMERIC

REAL

TEXT

NOT NULL

UNIQUE

PRIMARY KEY

FOREIGN KEY

JOIN

id	title
386676	The Office

show_id	rating
386676	…

title	id
The Office	386676

show_id	rating
386676	…

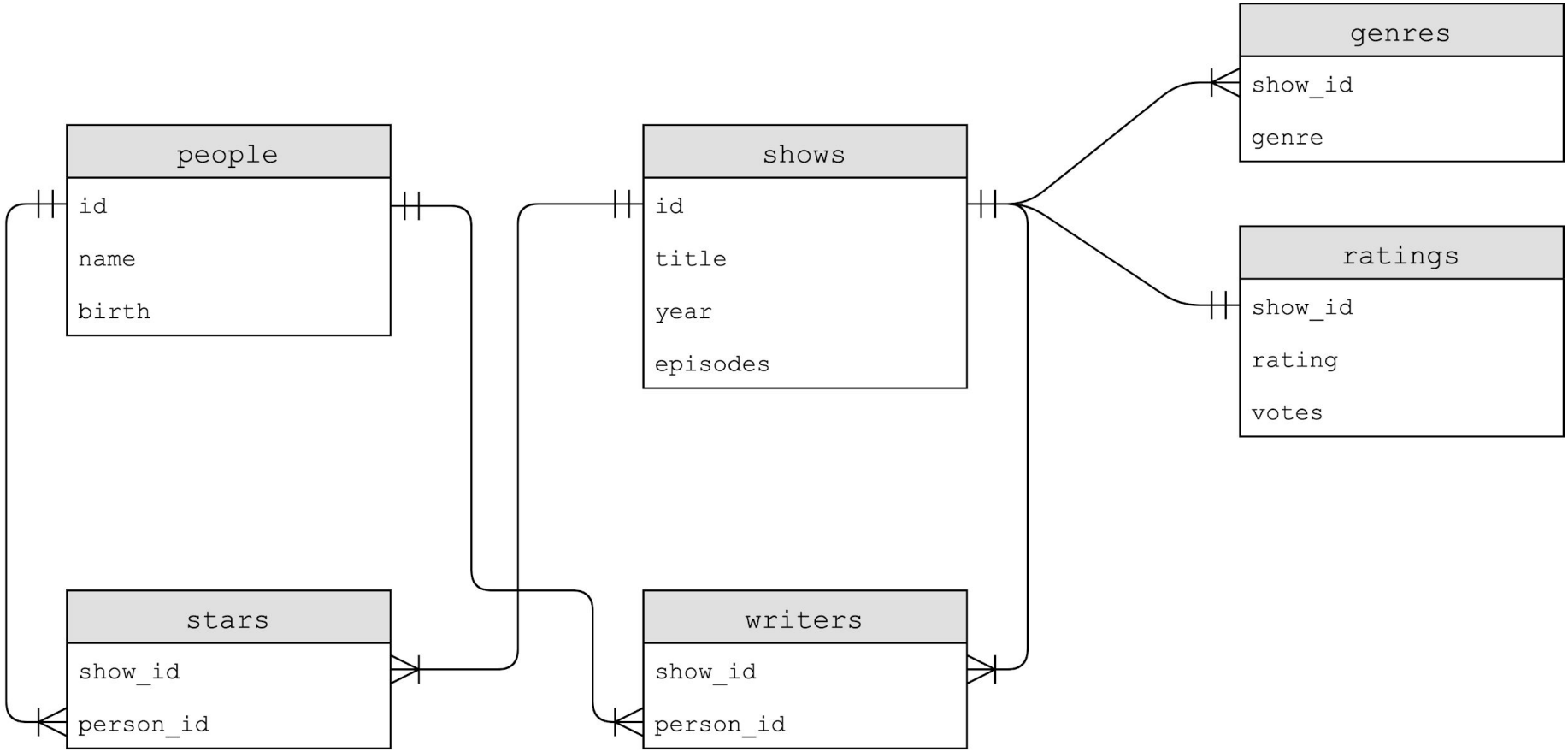
title	id
The Office	386676

show_id	rating
386676	…

title	id	show_id	rating
The Office	386676	386676	…

title	id	rating
The Office	386676	…

title	rating
The Office	…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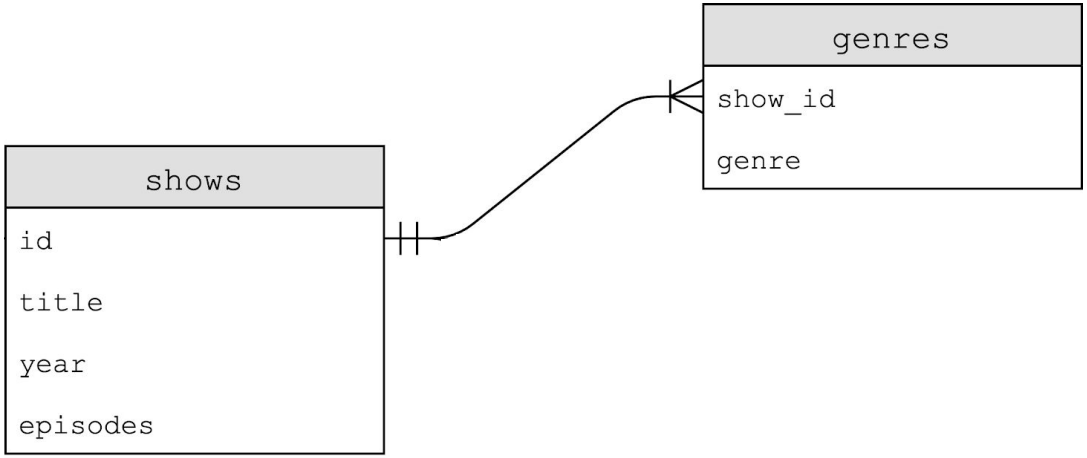
id
name
birth
shows
id
title
year
episodes
genres
show_id
genre
ratings
show_id
rating
votes
stars
show_id
person_id
writers
show_id
person_id


genres
show_id
genre
shows
id
title
year
episodes

one-to-many

id	title
63881	Catweazle

show_id	genre
63881	Adventure
63881	Comedy
63881	Family

title	id
Catweazle	63881

show_id	genre
63881	Adventure
63881	Comedy
63881	Family

title	id
Catweazle	63881

show_id	genre
63881	Adventure
63881	Comedy
63881	Family

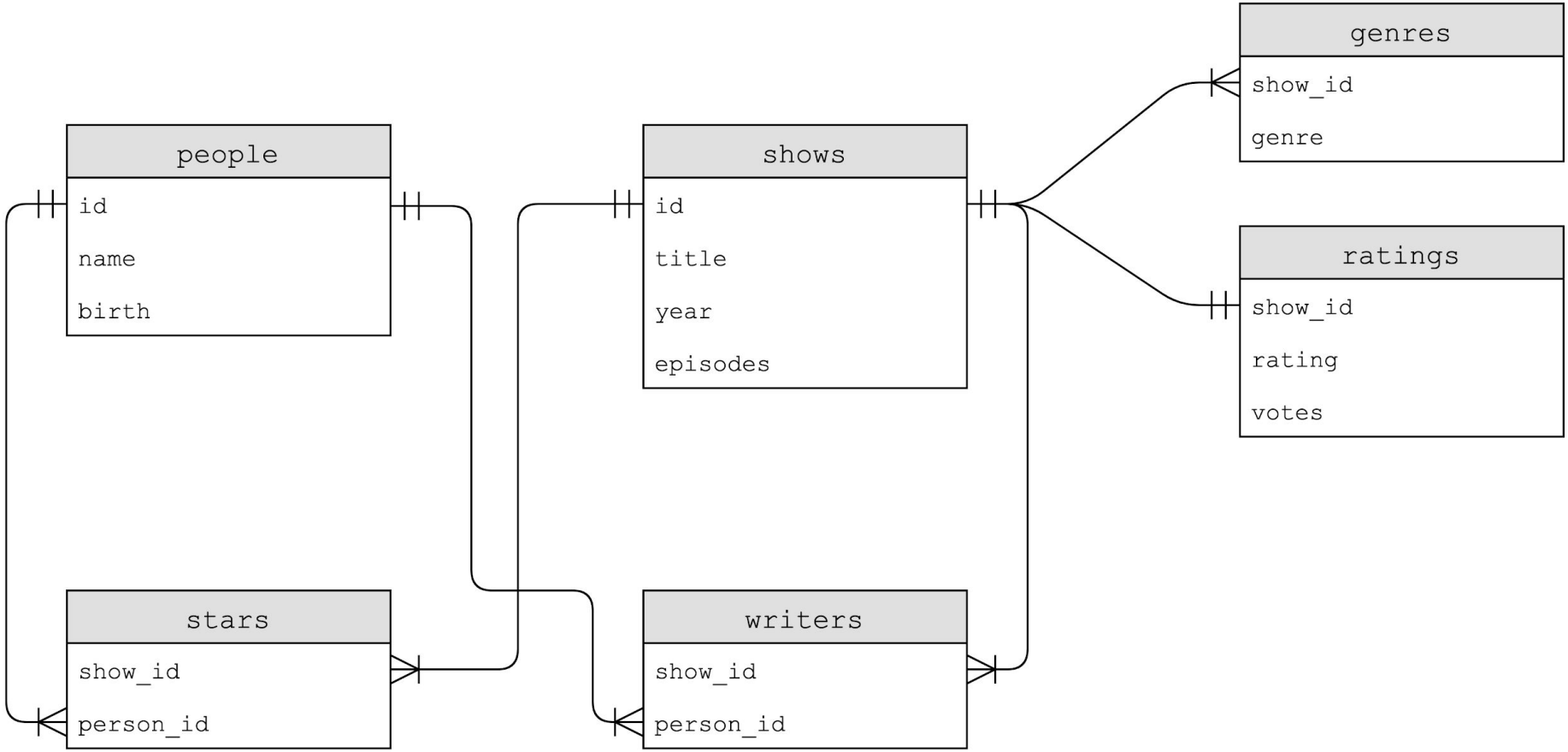
title	id	show_id	genre
Catweazle	63881	63881	Adventure
		63881	Comedy
		63881	Family

title	id	show_id	genre
Catweazle	63881	63881	Adventure
Catweazle	63881	63881	Comedy
Catweazle	63881	63881	Family

title	id	genre
Catweazle	63881	Adventure
Catweazle	63881	Comedy
Catweazle	63881	Family

title	genre
Catweazle	Adventure
Catweazle	Comedy
Catweazle	Family

genre
Adventure
Comedy
Family


people
id
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title
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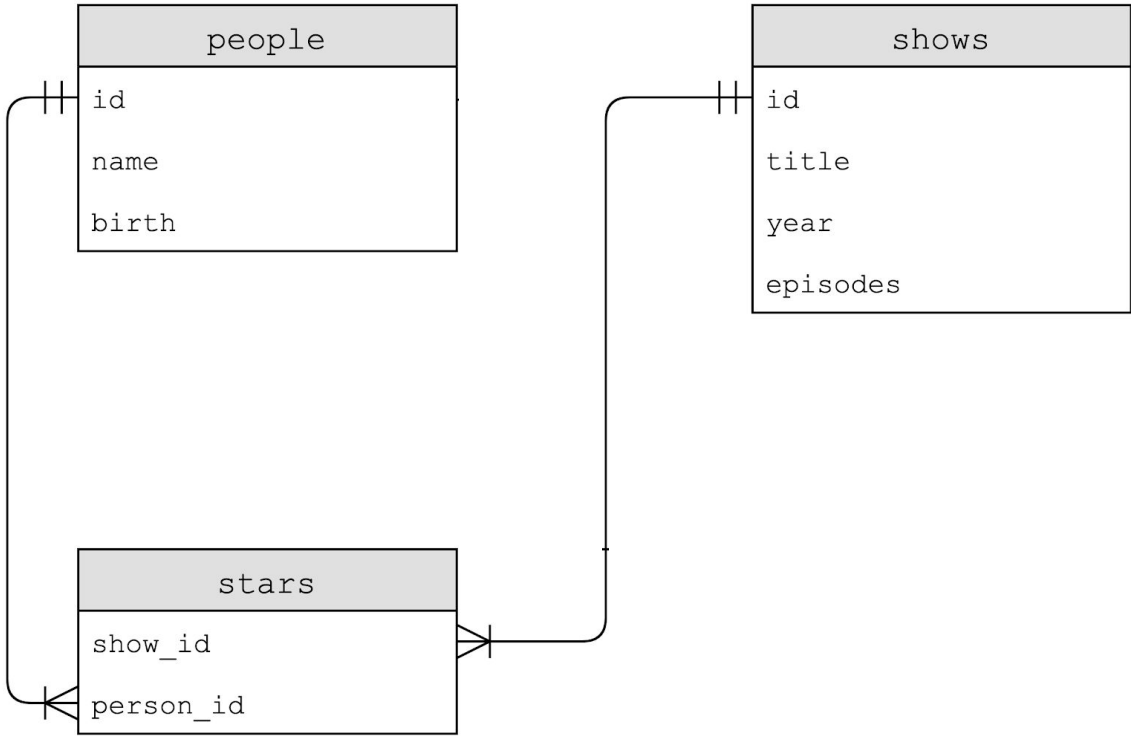
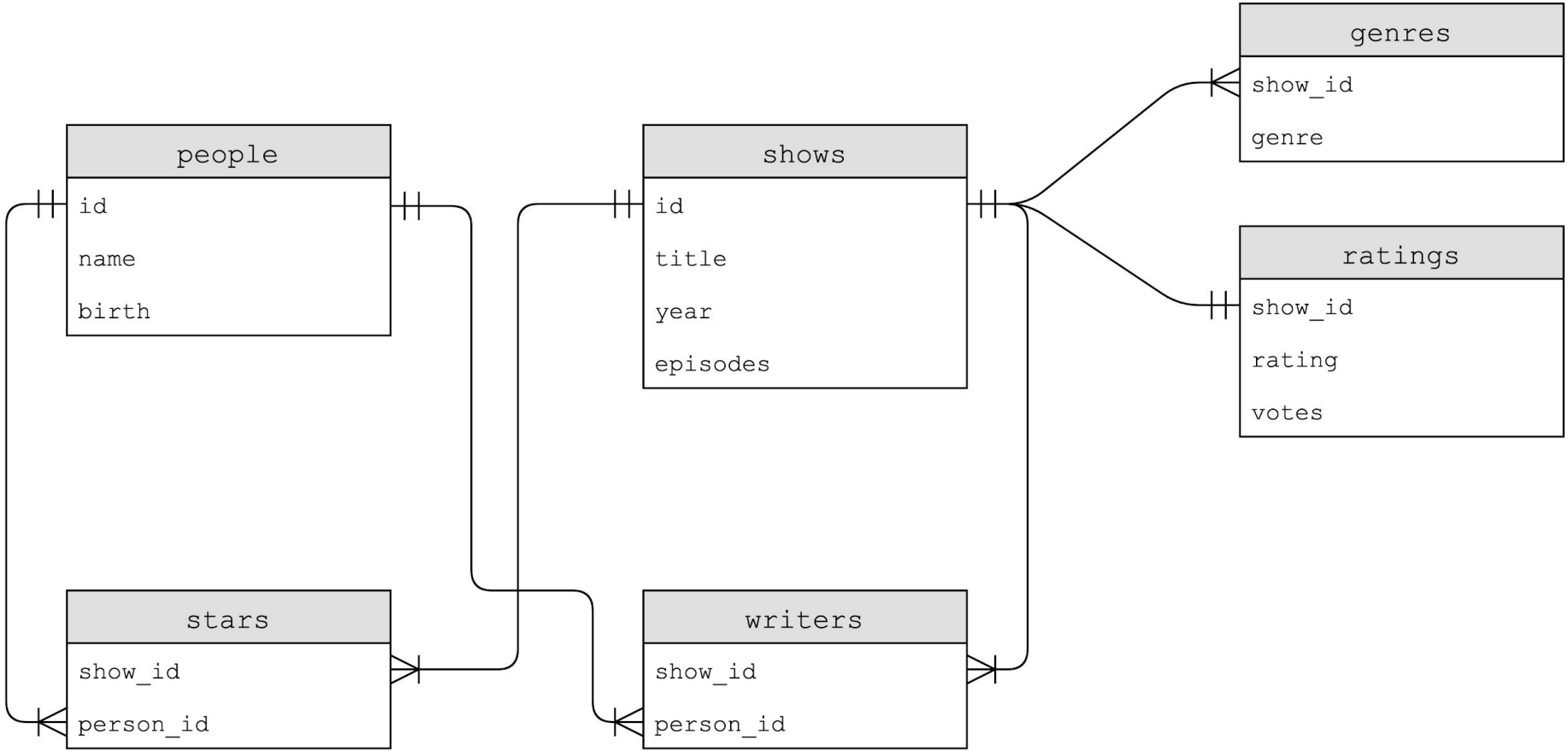
episodes
genres
show_id
genre
ratings
show_id
rating
votes
stars
show_id
person_id
writers
show_id
person_id


people
id
name
birth
shows
id
title
year
episodes
stars
show_id
person_id

many-to-many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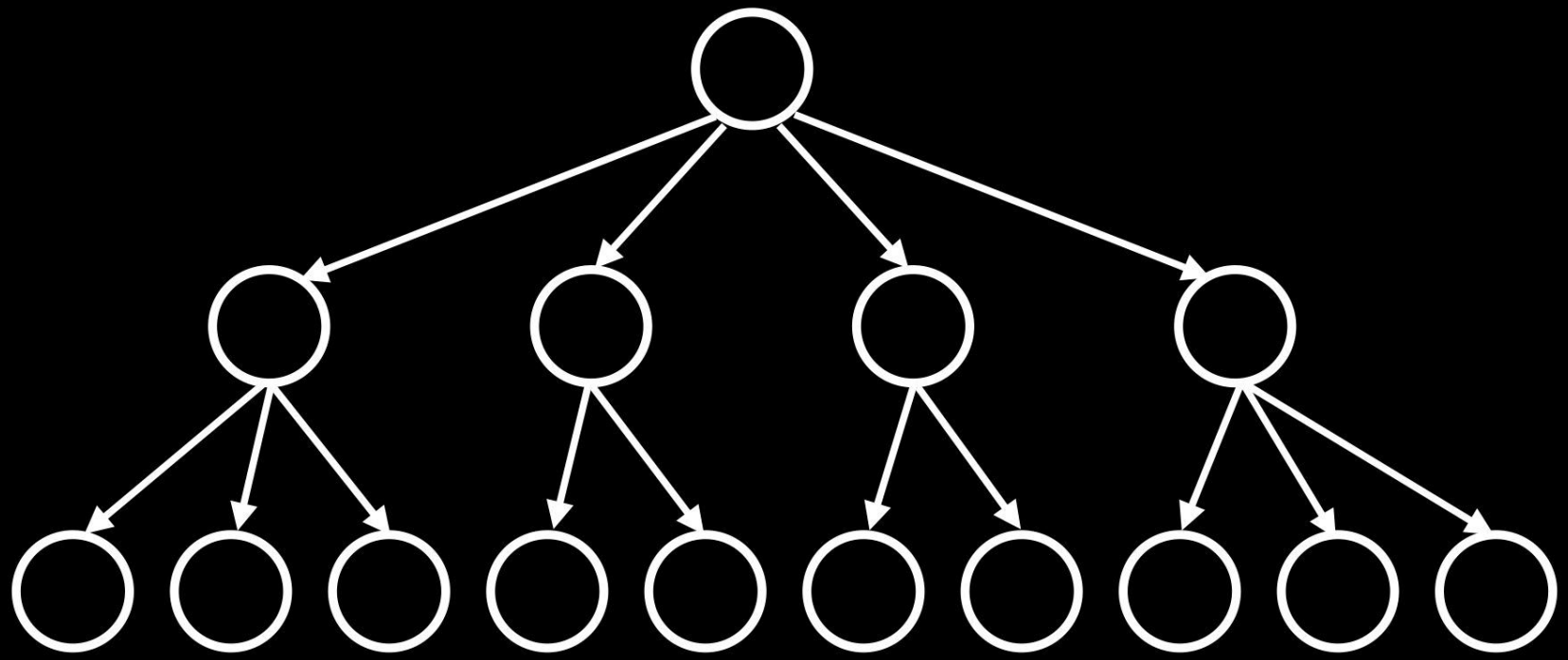
id
title
year
episodes
genres
show_id
genre
ratings
show_id
rating
votes
stars
show_id
person_id
writers
show_id
person_id

indexes

```
CREATE INDEX name ON table (column, ...);
```

B-trees



```
from cs50 import SQL
```

cs50.readthedocs.io/libraries/cs50/python/#cs50.SQL

SQL injection attacks

Sign in to GitHub

Username or email address

Password

Forgot password?

Sign in

Sign in to GitHub

Username or email address

malan@harvard.edu'--

Password

Forgot password?

Sign in

```
rows = db.execute("SELECT * FROM users WHERE username = ? AND password = ?", username, password)

if rows:
    ...
```

```
rows = db.execute("SELECT * FROM users WHERE username = ? AND password = ?", username, password)

if rows:
    ...
```

```
rows = db.execute("SELECT * FROM users WHERE username = ? AND password = ?", username, password)

if rows:
    ...
```

```
rows = db.execute(f"SELECT * FROM users WHERE username = '{username}' AND password = '{password}'")

if rows:
    ...
```

```
rows = db.execute(f"SELECT * FROM users WHERE username = 'malan@harvard.edu'--' AND password = '{password}'")

if rows:
    ...
```

```
rows = db.execute(f"SELECT * FROM users WHERE username = 'malan@harvard.edu'--' AND password = '{password}'")

if rows:
    ...
```

```
rows = db.execute(f"SELECT * FROM users WHERE username = 'malan@harvard.edu'--' AND password = '{password}'")

if rows:
    ...
```

```
rows = db.execute(f"SELECT * FROM users WHERE username = 'malan@harvard.edu''--' AND password = '{password}'")

if rows:
    ...
```

```
rows = db.execute("SELECT * FROM users WHERE username = ? AND password = ?", username, password)

if rows:
    ...
```

race conditions

Post

leomessi

75M 2M

```
rows = db.execute("SELECT likes FROM posts WHERE id = ?", id);
likes = rows[0]["likes"]
db.execute("UPDATE posts SET likes = ? WHERE id = ?", likes + 1, id);
```

BEGIN TRANSACTION

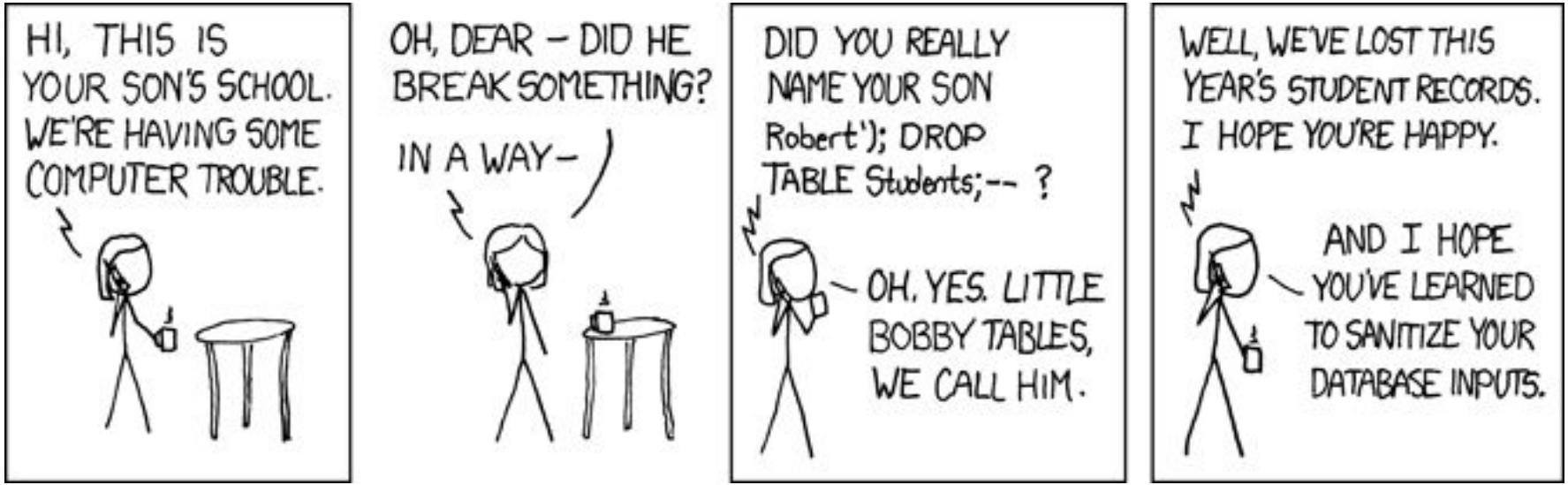
COMMIT

ROLLBACK

```
db.execute("BEGIN TRANSACTION")
rows = db.execute("SELECT likes FROM posts WHERE id = ?", id);
likes = rows[0]["likes"]
db.execute("UPDATE posts SET likes = ? WHERE id = ?", likes + 1, id);
db.execute("COMMIT")
```

ZU 0666', 0, 0); DROP DATABASE TABLICE;

ZU 0666', 0, 0); DROP DATABASE TABLICE;
PL


HI, THIS IS YOUR SON'S SCHOOL. WE'RE HAVING SOME COMPUTER TROUBLE.
OH, DEAR – DID HE BREAK SOMETHING?
IN A WAY–
DID YOU REALLY NAME YOUR SON Robert'); DROP TABLE Students;-- ?
OH. YES. LITTLE BOBBY TABLES, WE CALL HIM.
WELL, WE'VE LOST THIS YEAR'S STUDENT RECORDS. I HOPE YOU'RE HAPPY.
AND I HOPE YOU'VE LEARNED TO SANITIZE YOUR DATABASE INPUTS.

This is CS50